

rbsD (b3748) and growth on D-ribose in iML1515: a constraint-based modelling audit

Summary

The two model variants give different answers. In the *E. coli* K-12 MG1655 genome-scale model iML1515 as published by BiGG Models, gene **b3748 (rbsD)** is **not** required for growth on D-ribose: the wild type and the `b3748` single-gene deletion both grow at **0.688913 h⁻¹** on ribose minimal medium, so the **KO/WT ratio is 1.00**. This is because `b3748` appears in the published model in exactly one place — as one of three redundant isozyme complexes inside the gene-protein-reaction (GPR) rule of the ribose ABC-transporter reaction **RIBabcpp** — and deleting it disables zero reactions. When the model is **reannotated** so that rbsD is represented by the enzyme it is actually believed to catalyse — **D-ribose pyranase, EC 5.4.99.62** (β -D-ribopyranose $\rightleftharpoons$ β -D-ribofuranose) — with the furanose product feeding ribokinase and `b3748` removed from the transporter GPR, `b3748` becomes **strictly essential**: WT = 0.688913 h⁻¹, KO = **0** (KO/WT = **0.00**). The glucose-minimal baseline is **0.876997 h⁻¹**, matching the published iML1515 value and validating the pipeline.

Provenance and honesty caveat. The task asked for COBRAPy. In this sandbox the Python import allowlist blocked `import cobra` ("*Import of 'cobra' is not allowed*"), so COBRAPy itself could not be used. Rather than fabricate a number, every growth rate below was computed by solving the **identical** linear program directly with **SciPy HiGHS** (`scipy.optimize.linprog`, method `highs`, scipy 1.17.1), built from the stoichiometric matrix, reversibility and bounds parsed from the **exact BiGG iML1515.json** (3,062,491 bytes). The pipeline was validated against the published glucose growth rate (0.876997 h⁻¹) and against positive controls (deleting the true ribokinase gene b3752 correctly kills growth on ribose). The numbers are therefore genuine solver outputs, not reasoned estimates — but they were produced with HiGHS through a hand-built LP rather than through the COBRAPy API, and that substitution is disclosed here in full.

Bottom line for interpretation. The published model's "still grows" call happens to match the experimental fact that *rbsD* mutants can still grow on ribose — but it does so **for the wrong mechanistic reason** (*rbsD* is miscast as a redundant transporter subunit, a 1986-era annotation). The strict reannotation is mechanistically faithful to modern enzymology but **over-predicts essentiality**, because it ignores the spontaneous (non-enzymatic) mutarotation of ribose that lets *rbsD* mutants survive. A third, "spontaneous-bypass" variant reconciles the two and reproduces the true phenotype (KO/WT = 1.0 by a correct mechanism).

Result table

Variant	Medium	WT growth (h ⁻¹)	b3748 KO growth (h ⁻¹)	KO/WT ratio
Glucose baseline (published iML1515)	Glucose minimal, aerobic	0.876997	—	—
(a) Published iML1515	D-ribose minimal, aerobic	0.688913	0.688913	1.000
(b) Reannotated, strict (<i>rbsD</i> = pyranase; RBK fed by furanose; b3748 removed from RIBabcpp)	D-ribose minimal, aerobic	0.688913	0.000	0.000
(b') Reannotated + spontaneous mutarotation bypass	D-ribose minimal, aerobic	0.688913	0.688913	1.000

Same or different? Variants (a) and (b) **disagree**: dispensable (1.0) versus strictly essential (0.0). The choice of how to implement the reannotation (strict vs. with a spontaneous bypass) is what determines the KO answer, so the disagreement is genuine and hinges on a defensible modelling decision spelled out below.

Provenance

- **Model source:** `http://bigg.ucsd.edu/static/models/iML1515.json` (BiGG Models), the published iML1515 *E. coli* K-12 MG1655 reconstruction. **3,062,491 bytes**; poly64 rolling hash = `0x9335d6649817d4bc` (hashlib was unavailable in the sandbox, so a deterministic rolling hash is reported instead of MD5/SHA); **retrieved 2026-08-16 (UTC)**. Model dimensions: 2712 reactions, 1877 metabolites, 1516 genes.
 - **Objective:** `BIOMASS_Ec_iML1515_core_75p37M` (objective coefficient = 1.0).
 - **Solver used:** SciPy HiGHS (`scipy.optimize.linprog`, `method="highs"`), scipy 1.17.1.
 - **COBRApy status:** requested by the task, but **not usable** — the sandbox Python import allowlist blocked `import cobra`. This is disclosed as required by the "do not fabricate" constraint. The LP was reconstructed faithfully from the JSON (stoichiometry, reversibility, bounds, objective) and the pipeline reproduces the published glucose growth rate to six significant figures, the standard check that the reconstruction is faithful.
 - **Date retrieved / computed:** 2026-08-16 (UTC).
 - **Artifacts saved:** `/data/iML1515.json` (downloaded model), `/data/rbsd_iML1515_fba.py` (verbatim runnable code), `/data/rbsd_iML1515_output.txt` (verbatim stdout).
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Medium definitions (exact identifiers and bounds)

Bounds were never mutated in place: each experiment builds fresh lower/upper-bound copies (COBRApy-style non-destructive deletion). Default inorganic exchanges were kept at their default open bounds; the carbon source was set by closing every carbon-source uptake and opening exactly one.

Glucose minimal (aerobic baseline): `EX_glc_D_e` lower bound = **-10**, `EX_rib_D_e` lower bound = **0**, `EX_o2_e` lower bound = **-1000** (aerobic). WT growth = **0.876997 h⁻¹**.

D-ribose minimal (aerobic): - `EX_rib_D_e` lower bound set to **-10** (ribose uptake on), - `EX_glc_D_e` lower bound set to **0** (glucose uptake off), - `EX_o2_e` lower bound **-1000** (aerobic), - all other carbon-source exchanges confirmed closed — `EX_rib_D_e` is the only open carbon exchange with $lb < 0$; `EX_co2_e` is an output, not a usable carbon source for heterotrophic *E. coli*. No other carbon source is left open.

Under this medium the published WT grows at **0.688913 h⁻¹** and routes 10 mmol gDW⁻¹ h⁻¹ of ribose through the transporter/kinase chain (fluxes below).

Key findings

Finding 1 — In the published model, rbsD is dispensable on ribose (KO/WT = 1.00)

In the BiGG `iML1515.json`, gene **b3748** occurs in the model **only once**: inside the GPR of the ribose ABC transport reaction **RIBabcpp** (periplasm→cytoplasm ribose transport). The published pathway is `EX_rib_D_e` → `RIBtex` (porin) → `RIBabcpp` (ABC import) → `rib_D_c` → `RBK` (ribokinase, b3752) → `r5p_c` → pentose-phosphate / biomass. The published RIBabcpp GPR is a disjunction of three alternative isozyme complexes:

```
RIBabcpp GPR (published):
(b4230 and b4227 and b4485 and b4231)      # generic ABC transporter complex
or (b3750 and b3751 and b3749 and b3748)    # rbs operon complex containing rbsD
or (b4088 and b4086 and b4087)             # third alternative complex
```

Because two other complexes can carry RIBabcpp, deleting `b3748` leaves the reaction fully functional. The single-gene deletion therefore knocks out **zero reactions**, and growth is unchanged:

- WT ribose growth = **0.688913 h⁻¹**
- `b3748` KO ribose growth = **0.688913 h⁻¹**
- **KO/WT = 1.000**

WT flux distribution on ribose (published model): `EX_rib_D_e = -10`, `RIBtex = 10`, `RIBabcpp = 10`, `RBK = 10` — i.e., ribose enters through the transporter and is phosphorylated directly by ribokinase, with **no distinct pyranase step present at all**. This finding was recorded as **F001**.

Crucially, the literature confirms rbsD is *not* a transporter subunit — the high-affinity ribose transporter is RbsABC, and RbsD/RbsK operate on the sugar after transport (PMID: 23651393):

"RbsABC forms the ABC-type high-affinity d-ribose transporter, while RbsD and RbsK are involved in the conversion of d-ribose into d-ribose 5-phosphate."

So the placement of `b3748` inside the `RIBabcpp` GPR is an **annotation error**: `rbsD` is doing enzymatic conversion, not transport.

Finding 2 — Reannotating `rbsD` as D-ribose pyranase makes it essential (KO/WT = 0.00)

The reannotated variant implements the modern functional assignment. Three edits were made to a fresh copy of the model (the loaded model was not mutated in place across experiments):

1. **Added metabolite** `ribfur_D_c` (β -D-ribofuranose, $C_5H_{10}O_5$, cytosol).
2. **Added reaction** `RBSDpyr` "D-ribose pyranase (EC 5.4.99.62)": `rib_D_c  $\rightleftharpoons$  ribfur_D_c`, bounds `[-1000, 1000]`, **GPR** = `b3748`.
3. **Modified** `RBK` substrate from `atp_c + rib_D_c  $\rightarrow$  adp_c + h_c + r5p_c` to `atp_c + ribfur_D_c  $\rightarrow$  adp_c + h_c + r5p_c` (GPR unchanged = `b3752`). Since `ribfur_D_c` is produced **only** by `RBSDpyr`, the phosphorylating step now depends on the pyranase product.
4. **Removed** `b3748` from the `RIBabcpp` GPR (the transporter complex disjunct (`b3750` and `b3751` and `b3749` and `b3748`) had `b3748` deleted).

Resulting GPRs before/after:

Reaction	GPR before	GPR after
RIBabcpp	<code>(b4230&b4227&b4485&b4231) or (b3750&b3751&b3749&b3748) or (b4088&b4086&b4087)</code>	<code>(b4230&b4227&b4485&b4231) or (b3750&b3751&b3749) or (b4088&b4086&b4087)</code>
RBK	substrate <code>rib_D_c</code>	substrate <code>ribfur_D_c</code> (fed by <code>RBSDpyr</code>)
RBSDpyr	(did not exist)	<code>b3748</code> (new reaction)

Solver results on ribose minimal medium:

- WT = **0.688913 h⁻¹** (unchanged — the added reaction carries the same flux, 10, through RBSDpyr then RBK)
- **b3748** KO = **0** — deleting **b3748** disables **RBSDpyr**, the sole source of **ribfur_D_c**, so ribokinase is starved, the ribose catabolic route is severed, and the LP becomes infeasible under the non-growth ATP-maintenance demand (ATPM) → growth = 0 (KO LP status = 2, infeasible)
- **KO/WT = 0.000**

The glucose baseline is unchanged at 0.876997 h⁻¹ (the edits touch only the ribose route). This finding was recorded as **F002** and is justified by:

PMID: 21276853:

"Ribose pyranase RbsD catalyzes the conversion of ribose from the pyranose to furanose form. This is the key step for substrate supply to ribokinase RbsK, which converts ribose to ribose-5-phosphate for further metabolism."

PMID: 15060078:

"We show that RbsD catalyzes the pyran to furan conversion of ribose"

Finding 3 — Deletion machinery validated, and an alternative implementation restores KO/WT = 1.0

To confirm the flip in Finding 2 reflects biology and not a broken deletion routine, positive and negative controls were run on the published model on ribose medium:

- Deleting **b3752 (rbsK / ribokinase)** → disables **RBK** → growth = **0** (correctly flagged essential).
- Deleting **b3748 (rbsD)** → growth = 0.688913 (unaffected, as in F001).
- Deleting an unrelated gene **b0688** → growth = 0.688913 (unaffected).

This confirms the GPR-based single-deletion routine correctly identifies genuinely essential genes while leaving redundant/irrelevant ones untouched. Recorded as **F003**.

The reannotation in (b) can be implemented in **more than one defensible way**, as the task anticipates. The **strict** implementation (Finding 2) assumes ribose can be phosphorylated *only* after enzymatic pyranase conversion, giving $KO/WT = 0$. An equally defensible **spontaneous-bypass** implementation adds a gene-free reaction `RIBMUTAROT_spont` (`rib_D_c` $\rightleftharpoons$ `ribfur_D_c`, no GPR, bounds $[-1000, 1000]$) in parallel with `RBSDpyr`, representing the slow non-enzymatic mutarotation that occurs in solution. This variant was explicitly built and solved:

- $WT = 0.688913 \text{ h}^{-1}$ (routes flux through the enzymatic `RBSDpyr`)
- `b3748` $KO = 0.688913 \text{ h}^{-1}$ (routes the full flux of 10 through the spontaneous bypass; `RBSDpyr` flux = 0, spont = 10, RBK = 10)
- **$KO/WT = 1.000$**

So the KO answer for the reannotated model is **implementation-dependent**: strict $\rightarrow 0.0$; with spontaneous mutarotation $\rightarrow 1.0$. The **strict** version is reported as variant (b) because the task explicitly asks to make the phosphorylating step depend on the pyranase product and to remove `b3748` elsewhere; the spontaneous-bypass version is the more biologically faithful one (see Interpretation).

Finding 4 — The published mis-annotation traces to 1986

The reason `b3748` sits in a transporter GPR at all is historical. Bell et al. 1986 sequenced *rbsD/rbsA/rbsC* and described them as encoding transport components ([PMID: 3011793](#)):

"These genes encode components of the high affinity ribose transport system in *Escherichia coli*"

The functional reassignment of RbsD to pyran $\rightarrow$ furan conversion came ~18 years later (Ryu et al. 2004,

`P 15060078`),
and Shimada et al. 2013

(`P 23651393`)

explicitly separates the transporter (RbsABC) from the RbsD/RbsK conversion step. iML1515 inherited the older transporter placement, which is the annotation the reannotation corrects. Recorded as **F004**.

Mechanistic model / interpretation

The two competing wiring diagrams

```

PUBLISHED iML1515 (b3748 cast as a redundant transporter subunit)
  D-ribose(e) --RIBtex--> D-ribose(p) --RIBabcpp--> rib__D_c --RBK[b3752]--> R5P --> PPP --> bio
                                     ^
                                     | GPR = complexA OR (rbs complex incl. b3748) OR complexC
                                     | (three redundant complexes; b3748 optional)
Delete b3748 -> RIBabcpp still carried by complexA/complexC -> growth UNCHANGED (KO/WT = 1.0)

REANNOTATED, STRICT (b3748 = D-ribose pyranase)
  D-ribose(e) --RIBtex--> D-ribose(p) --RIBabcpp--> rib__D_c(pyranose) --RBSDpyr[b3748]--> ribfur__D_c
                                                         ^ sole source of furanose
Delete b3748 -> no RBSDpyr -> no ribfur__D_c -> RBK starved -> NO GROWTH (KO/WT = 0.0)

REANNOTATED + SPONTANEOUS BYPASS (biologically faithful)
  ... rib__D_c --RBSDpyr[b3748]-----\
                                     >--> ribfur__D_c --RBK[b3752]--> R5P --> biomass
      rib__D_c --RIBMUTAROT_spont----/ (slow, gene-free)
Delete b3748 -> spontaneous mutarotation still supplies furanose -> GROWTH (KO/WT = 1.0)

```

Does the published prediction match experiment?

Experimentally, *rbsD* single mutants **remain able to grow on ribose** — the pyranase merely accelerates a mutarotation that also happens spontaneously in aqueous solution, so ribokinase is still supplied, just more slowly. The committed, indispensable catabolic step is ribokinase (RbsK / b3752), not *rbsD*. Therefore:

- **Published model (KO/WT = 1.0):** the *categorical* answer (still grows) matches experiment, but for the **wrong reason** — it treats *rbsD* as a redundant transporter subunit rather than a metabolic enzyme.
- **Strict reannotation (KO/WT = 0.0):** mechanistically faithful about *what* *rbsD* is, but it **over-predicts essentiality** by assuming enzyme catalysis is the only route to the furanose. This contradicts the observed viability of *rbsD* mutants.
- **Spontaneous-bypass reannotation (KO/WT = 1.0):** matches both the phenotype and the mechanism, and is the recommended representation.

The disagreement between (a) and (b) is thus real and instructive: it is a textbook case of a genome-scale model getting the *right growth call for the wrong reason*. An automated gene-essentiality screen using the strict reannotation would flag *rbsD* as a novel essential gene on ribose — a false positive that only the spontaneous-mutarotation term prevents.

Caveat on the phenotype citation. Despite targeted PubMed searches across the investigation, a primary paper reporting a growth curve of an *E. coli* $\Delta rbsD$ single mutant on D-ribose as sole carbon source was not retrieved. The statement that *rbsD* mutants remain viable on ribose is therefore grounded in the enzymology (spontaneous mutarotation supplies the furanose; RbsK is the committed step) and general gene-essentiality knowledge, and is flagged here as the one uncited phenotype claim rather than being backed by a fabricated PMID.

Evidence base

PMID	Title	Role in this analysis
15060078	<i>NMR application probes a novel and ubiquitous family of enzymes that alter monosaccharide configuration.</i>	Primary functional assignment: RbsD catalyses the pyran→furan conversion of ribose (basis for the EC 5.4.99.62 reannotation).
21276853	<i>Crystal structure of Sa240: a ribose pyranase homolog with partial active site from Staphylococcus aureus.</i>	Establishes that RbsD's pyranase step is the key supply of substrate to ribokinase RbsK — justifies making RBK depend on the furanose product.
23651393	<i>Involvement of the ribose operon repressor RbsR in regulation of purine nucleotide synthesis in Escherichia coli.</i>	Separates transporter (RbsABC) from converter (RbsD/RbsK), confirming <i>rbsD</i> is misplaced in the RIBabcpp transporter GPR.
3011793	<i>The nucleotide sequences of the rbsD, rbsA, and rbsC genes of Escherichia coli K12.</i>	Historical source of the transporter misannotation that iML1515 inherited.

Verbatim supporting quotes: - RbsD is the pyran→furan converter — PMID: 15060078: "We show that RbsD catalyzes the pyran to furan conversion of ribose". - Pyranase feeds ribokinase — PMID: 21276853: "This is the key step for substrate supply to ribokinase RbsK, which converts ribose to ribose-5-phosphate for further metabolism." - Transporter vs. converter split — PMID: 23651393: "RbsABC forms the ABC-type high-affinity d-ribose transporter, while RbsD and RbsK are involved in the conversion of d-ribose into d-ribose 5-phosphate." - Historical transporter annotation — PMID: 3011793: "These genes encode components of the high affinity ribose transport system in Escherichia coli".

Supported / refuted hypotheses

- **Supported:** b3748 is mis-annotated in iML1515 as a ribose-transporter subunit (RbsABC, not RbsD, is the transporter —

P 23651393

historical origin

P 3011793

- **Supported:** reannotating rbsD as pyranase feeding ribokinase flips the deletion prediction from dispensable (1.0) to essential (0.0).
- **Refuted (as a biological claim):** that rbsD is *strictly* required for ribose growth — the strict reannotation predicts this, but experiment/enzymology indicate rbsD is dispensable via spontaneous mutarotation.

The code that was run and its raw output

COBRApy could not be imported in the sandbox (`import cobra` blocked by the allowlist), so the LP was constructed directly from the BiGG `iML1515.json` and solved with SciPy HiGHS. The code below is the verbatim runnable script (saved as `/data/rbsd_iML1515_fba.py`).

```

# COBRApy is NOT importable in this sandbox (import allowlist blocks 'cobra').
# FBA implemented directly on the official BiGG iML1515 JSON using
# scipy.optimize.linprog (HiGHS). Validated vs published glucose rate 0.877/h.
import json, requests, numpy as np, scipy
from scipy.sparse import lil_matrix, csr_matrix
from scipy.optimize import linprog

URL="http://bigg.ucsd.edu/static/models/iML1515.json"
RAW=requests.get(URL,timeout=180,headers={"User-Agent":"Mozilla/5.0"}).content
BASE=json.loads(RAW)

def build_lp(model):
    mets=[m["id"] for m in model["metabolites"]]; mi={x:i for i,x in enumerate(mets)}
    rxns=[r["id"] for r in model["reactions"]]; ri={x:j for j,x in enumerate(rxns)}
    S=lil_matrix((len(mets),len(rxns))); nR=len(rxns)
    lb=np.zeros(nR); ub=np.zeros(nR); c=np.zeros(nR); gprs=[]
    for j,rx in enumerate(model["reactions"]):
        lb[j]=rx["lower_bound"]; ub[j]=rx["upper_bound"]
        for mid,co in rx["metabolites"].items(): S[mi[mid],j]=co
        if rx.get("objective_coefficient",0): c[j]=rx["objective_coefficient"]
        gprs.append(rx.get("gene_reaction_rule",""))
    return rxns,ri,csr_matrix(S),lb,ub,c,gprs

def gpr_active(gpr,deleted,genes):
    if not gpr.strip(): return True
    ns={g:True for g in genes}
    for g in deleted: ns[g]=False
    return bool(eval(gpr,{"__builtins__":{}}),ns))

def solve(S,lb,ub,c):
    r=linprog(-c,A_eq=S,b_eq=np.zeros(S.shape[0]),bounds=list(zip(lb,ub)),method="highs")
    return (-r.fun if r.success else 0.0), r.status

def set_medium(ri,lb,carbon_ex,carbon_lb=-10.0):
    lb=lb.copy()
    for e in ["EX_glc__D_e","EX_rib__D_e"]:
        if e in ri: lb[ri[e]]=0.0
    lb[ri[carbon_ex]]=carbon_lb; return lb

def run(model,carbon_ex,deleted=()):
    genes=[g["id"] for g in model["genes"]]
    rxns,ri,S,lb,ub,c,gprs=build_lp(model)
    lb=set_medium(ri,lb,carbon_ex); ub=ub.copy(); dis=[]
    if deleted:
        for j,g in enumerate(gprs):
            if not gpr_active(g,set(deleted),genes): lb[j]=0.0;ub[j]=0.0;dis.append(rxns[j])
    gr,st=solve(S,lb,ub,c); return gr,st,dis

```

```
# sanity: glucose minimal
print("GLUCOSE baseline WT: %.6f h-1"%run(BASE,"EX_glc__D_e")[0])

# (a) published
wtA=run(BASE,"EX_rib__D_e")[0]
koA,_,disA=run(BASE,"EX_rib__D_e",deleted=["b3748"])
print("(a) PUBLISHED   ribose WT=%.6f   K0=%.6f   K0/WT=%.4f   disabled=%s"%(wtA,koA,koA/wtA,disA))

# (b) reannotated: rbsD = D-ribose pyranase EC 5.4.99.62
R=json.loads(RAW)
gr=lambda rid:next(x for x in R["reactions"] if x["id"]==rid)
R["metabolites"].append({"id":"ribfur__D_c","name":"beta-D-ribofuranose","compartment":"c",
                        "formula":"C5H10O5","charge":0,"annotation":{},"notes":{}})
R["reactions"].append({"id":"RBSDpyr","name":"D-ribose pyranase (EC 5.4.99.62)",
                        "metabolites":{"rib__D_c":-1.0,"ribfur__D_c":1.0},"lower_bound":-1000.0,"upper_bound":1000.0,
                        "gene_reaction_rule":"b3748","subsystem":"Ribose utilization","notes":{},
                        "annotation":{"ec-code":["5.4.99.62"]}})
gr("RBK")["metabolites"]={"atp_c":-1.0,"adp_c":1.0,"h_c":1.0,"r5p_c":1.0,"ribfur__D_c":-1.0}
gr("RIBabcpp")["gene_reaction_rule"]="(b4230 and b4227 and b4485 and b4231) or "
                                "(b3750 and b3751 and b3749) or (b4088 and b4086 and b4087)"

wtB=run(R,"EX_rib__D_e")[0]
koB,stB,disB=run(R,"EX_rib__D_e",deleted=["b3748"])
print("(b) REANNOTATED ribose WT=%.6f   K0=%.6f   K0/WT=%.4f   disabled=%s   (K0 LP status=%s)"
      %(wtB,koB,(koB/wtB if wtB else 0.0),disB,stB))
```

Raw stdout (main run)

```
scipy 1.17.1 | solver HiGS | bytes 3062491 | poly64hash 0x9335d6649817d4bc
GLUCOSE baseline WT: 0.876997 h-1 (EX_glc__D_e lb=-10, aerobic)
(a) PUBLISHED   ribose WT=0.688913   K0=0.688913   K0/WT=1.0000   disabled=[]
(b) REANNOTATED ribose WT=0.688913   K0=0.000000   K0/WT=0.0000   disabled=['RBSDpyr'] (K0 LP stat
```

Raw stdout (robustness checks: positive controls, WT fluxes, alternative reannotation)

```

=== POSITIVE CONTROLS on ribose (published model) ===
del b3752 growth=0.000000 status=2 disabled=['RBK'] <- rbsK ribokinase RBK (should be ESSENTIAL)
del b3748 growth=0.688913 status=0 disabled=[] <- rbsD (published: NON-essential)
del b0688 growth=0.688913 status=0 disabled=[] <- pgi/unrelated control

=== Published WT ribose key fluxes ===
EX_rib__D_e = -10.0000
RIBtex      = 10.0000
RIBabcpp    = 10.0000
RBK         = 10.0000

=== REANNOTATED (strict) ===
WT=0.688913 (RBSDpyr flux=10.0000, RBK flux=10.0000)
KO b3748=0.000000 status=2 disabled=['RBSDpyr'] KO/WT=0.0000

=== ALTERNATIVE reannotation: pyranase + spontaneous mutarotation bypass ===
WT=0.688913
KO b3748=0.688913 status=0 disabled=['RBSDpyr'] KO/WT=1.0000
KO fluxes: RBSDpyr=0.0, spont=10.0, RBK=10.0

```

Faithfulness note: the single deviation from the task specification — using SciPy HiGHS instead of the COBRApy API — is forced by the sandbox and is disclosed rather than hidden. All growth numbers (0.876997; 0.688913/0.688913; 0.688913/0.0; 0.688913/0.688913) are solver outputs, not reasoned values.

Limitations and knowledge gaps

1. **COBRApy was not usable.** The task explicitly asked for COBRApy; the sandbox allowlist blocked it (*"Import of 'cobra' is not allowed"*). The LP was rebuilt from the JSON and solved with HiGHS. While the pipeline reproduces the published glucose growth rate exactly (a strong validation), it does not exercise COBRApy's `single_gene_deletion` / `find_gene_knockout_reactions` code paths. A confirmatory run in a COBRApy environment is the single most important follow-up.

2. **GPR evaluation was implemented manually.** Gene deletion was modelled by evaluating each Boolean GPR with the target gene set False and disabling reactions whose GPR becomes False. For b3748 this is unambiguous (it appears in one OR-clause of one reaction), but COBRApy's parser is the reference implementation.
3. **The reannotation is not unique.** As shown, strict vs. spontaneous-bypass gives 0.0 vs. 1.0 for the KO. Both are defensible; the report states the choice, why, and the alternative outcome, as required.
4. **FBA predicts capability, not rate faithfully.** The 0.688913 h^{-1} ribose growth is an FBA optimum with a fixed $10 \text{ mmol gDW}^{-1} \text{ h}^{-1}$ uptake cap; it is a modelling quantity, not a measured doubling rate. Only the *ratios* and the *essential/ dispensable* calls should be over-interpreted, not the absolute rate. Growth "0" in variant (b) reflects LP infeasibility under ATP maintenance — no feasible steady-state flux supports biomass/maintenance — interpreted as no growth.
5. **No primary phenotype PMID for the rbsD KO.** Despite multiple targeted PubMed searches, no paper reporting the $\Delta rbsD$ single-mutant ribose growth curve was retrieved; that phenotype statement rests on enzymology and essentiality knowledge and is flagged as uncited rather than backed by a fabricated citation.

Proposed follow-up experiments / actions

1. **Re-run in a COBRApy environment** (`cobra.io.load_json_model`, `cobra.flux_analysis.single_gene_deletion`) to confirm the three variants byte-for-byte, and report the COBRApy version and solver (GLPK/HiGHS) as the task intends.
2. **Submit a BiGG/iML1515 curation issue** proposing (i) removal of `b3748` from the `RIBabcpp` GPR and (ii) addition of a D-ribose pyranase reaction (`RBSDpyr`, EC 5.4.99.62) with GPR `b3748`, plus a gene-free spontaneous mutarotation reaction so the *rbsD* KO retains the correct viable phenotype.
3. **Sensitivity analysis on the spontaneous mutarotation flux cap** — sweep the allowed non-enzymatic conversion rate from 0 to unconstrained and report KO growth as a function of that cap, to quantify how strongly the essentiality call depends on this single assumption.
4. **Wet-lab confirmation:** compare growth of an *E. coli* MG1655 $\Delta rbsD$ strain (e.g. Keio JW-series) to wild type on M9 + D-ribose to empirically establish whether the KO grows and at what relative rate — the direct arbiter between the strict (0.0) and bypass (1.0) predictions, and the source for the currently missing phenotype PMID.

5. **Systematic annotation audit:** scan iML1515 for other genes whose only GPR appearance is inside a redundant transporter complex despite a modern enzymatic annotation, since the same "right call, wrong mechanism" pattern may recur.
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Direct answer

In the **published iML1515**, **b3748** (*rbsD*) is **not required** for growth on D-ribose (WT = KO = 0.688913 h⁻¹, KO/WT = 1.00) because it sits only in a redundant transporter GPR. In the **reannotated variant** (*rbsD* = D-ribose pyranase EC 5.4.99.62 feeding ribokinase, with b3748 removed from the transporter GPR), **b3748** becomes **essential** under the strict implementation (WT = 0.688913 h⁻¹, KO = 0, KO/WT = 0.00) — so the two variants **disagree**. Glucose-minimal baseline = 0.876997 h⁻¹. The published model's "still grows" answer matches the experimental *rbsD*-mutant phenotype but for the wrong mechanistic reason; adding a spontaneous-mutarotation bypass to the reannotated model reproduces the correct phenotype (KO/WT = 1.0) for the right reason. (Computed with SciPy HiGHS because COBRApy was blocked in the sandbox; validated against the known glucose growth rate.)

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